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Aero 4720

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Test 2: for a trapezoidal integration scheme, show:

1. The resultant force F over the interval S is $F = \frac{S(a+b)}{2}$

$$F = \int_0^S b - (b-a) \left(\frac{x}{S} \right) dx$$

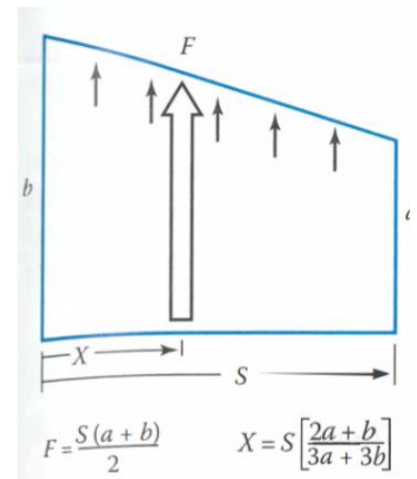
$$F = bx - (b-a) \left(\frac{x^2}{2S} \right) \Big|_0^S$$

$$bS - (b-a) \left(\frac{S^2}{2S} \right) - 0$$

$$bS - \frac{bS - aS}{2}$$

$$\frac{bS + aS}{2}$$

$$\frac{S(a+b)}{2}$$



2. The resultant force's location x

$$X = \frac{\int_0^S \left(b - (b-a) \left(\frac{x}{S} \right) \right) x dx}{F} = \frac{2}{S(a+b)} \int_0^S \left(b - (b-a) \left(\frac{x}{S} \right) \right) x dx$$

$$X = \frac{2}{S(a+b)} \int_0^S bx - (b-a) \left(\frac{x^2}{S} \right) dx$$

$$X = \frac{2}{S(a+b)} \left[\frac{bx^2}{2} - (b-a) \left(\frac{x^3}{3S} \right) \right] \Big|_0^S$$

$$X = \frac{2}{S(a+b)} \left[\frac{bS^2}{2} - \left(\frac{bS^3}{3S} - \frac{aS^3}{3S} \right) \right]$$

$$X = \frac{2}{S(a+b)} \left[\frac{bS^2}{2} - \left(\frac{bS^2}{3} - \frac{aS^2}{3} \right) \right]$$

$$X = \frac{2}{S(a+b)} \left[\frac{3bS^2}{6} - \left(\frac{2bS^2}{6} - \frac{2aS^2}{6} \right) \right]$$

$$X = \frac{2}{S(a+b)} \left[\frac{bS^2}{6} + \frac{2aS^2}{6} \right]$$

$$X = \frac{S^2}{S(a+b)} \left[\frac{b}{3} + \frac{2a}{3} \right]$$

$$X = \frac{S}{(a+b)} \frac{b+2a}{3}$$

$$X = S \frac{b+2a}{3a+3b}$$